

# Theoretical Bounds on $\chi_\nu$ in Observer-Patch Holography

Canonical Coherent-Matter Susceptibility and Repair-Charge Coupling

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## Abstract

The coherent-material branch of observer-patch holography supplies a canonical scalar source in the same quotient-edge register as the protected collar reserve. On the co-registered presence branch, the dimensionless coupling is  $\chi_\nu^{\text{can}} = 1 - P/24 = 0.9320429912748350\dots$ . Under the weaker mean-count reading it lies between that value and one. This is a channel coefficient, not a force coefficient. A proposed repair-charge condensate action couples the source to a dynamical repair phase. Conditional on that action, repair-current continuity, dilute dust behavior, spherical deep-galaxy scaling, field stress, and a lawful coherent-source force follow from one variational system. A compact neutral coherence contrast has no monopole and cannot support a closed device. The dimensional source calibration, ambient repair field, compact-phase completion, and laboratory source receipt are work in progress; no numerical lift prediction follows from  $\chi_\nu$  alone.

## Claim Boundary

This paper proves a finite-channel coefficient on a declared OPH branch. It also derives continuity, dilute dust behavior, spherical deep-galaxy scaling, force, and momentum formulae conditional on the repair-charge condensate action in the dark-sector paper [4]. The OPH derivation of that action and its dimensional source normalization are work in progress.

The exact statuses are:

| Object                                                    | Status                                                             |
|-----------------------------------------------------------|--------------------------------------------------------------------|
| Coherent source in the quotient-edge scalar register      | finite theorem under the stated source receipts                    |
| $\chi_\nu^{\text{can}} = \lambda_{\text{collar}}$         | theorem under co-registration and reserve-survival premises        |
| $\chi_\nu^{\text{can}} = 1 - P/24$                        | exact on the presence branch                                       |
| Repair-charge action                                      | proposed dynamical completion                                      |
| Continuity, dust limit, and spherical deep-galaxy scaling | derived conditional on that action and its constitutive premises   |
| Dimensional coupling $q_\star$                            | work in progress                                                   |
| Device force                                              | conditional on source charge, external field, and momentum closure |
| Numerical lift                                            | absent                                                             |

## Contents

|                                           |          |
|-------------------------------------------|----------|
| <b>1 Canonical Coherent-Matter Source</b> | <b>2</b> |
| <b>2 Protected-Reserve Coefficient</b>    | <b>3</b> |
| <b>3 Engineering Coordinates</b>          | <b>4</b> |
| <b>4 Repair-Charge Coupling</b>           | <b>4</b> |
| <b>5 Compact-Source Constraint</b>        | <b>6</b> |
| <b>6 Momentum and Energy Closure</b>      | <b>6</b> |
| <b>7 Laboratory Evidence Conditions</b>   | <b>7</b> |
| <b>8 Conclusion</b>                       | <b>7</b> |

## 1 Canonical Coherent-Matter Source

An OPH technology instantiation is an observer-like self-reading system: a bounded physical or software patch with local state, ports or boundaries, readback, records, feedback or repair moves, and a public evidence bundle. For a nondegenerate coherent material subfederation  $U$ , define the canonical source

$$S_{\text{coh}}^{\text{can}} = \mathbf{1}_{\text{self-read}} \mathbf{R}_U \mathbf{P}_U \mathbf{C}_U. \quad (1.1)$$

The factors state that the patch reads itself, produces a stable record, supports a predictive boundary, and executes a registered repair or control move. If any factor vanishes, the coherent scalar source is zero.

The screen-microphysics branch supplies four structural facts [3]:

- S1.  $S_{\text{coh}}^{\text{can}}$  descends to a protected quotient observable.
- S2. A nondegenerate source is nonzero.
- S3. Scalar edge-center exhaustion places it in the unique quotient-local scalar register.
- S4. The protected  $\mathbb{Z}_6$  reserve acts on that same register and commutes with scalar activation.

These statements identify the channel. They supply no spacetime action or dimensional gravitational charge.

## 2 Protected-Reserve Coefficient

Let  $\epsilon_{\mathbb{Z}_6}(y)$  be the conditional reserve-presence profile on the co-registered scalar slot, and let  $w(y)$  be the normalized scalar weight across the finite collar. The scalar opportunity survives with local factor

$$\lambda_{\text{slot}}(y) = 1 - \epsilon_{\mathbb{Z}_6}(y). \quad (2.1)$$

The finite-thickness collar coefficient is

$$\lambda_{\text{collar}} = \int dy w(y) [1 - \epsilon_{\mathbb{Z}_6}(y)]. \quad (2.2)$$

The protected-reserve receipt gives

$$\int dy w(y) \epsilon_{\mathbb{Z}_6}(y) = \frac{P}{24}. \quad (2.3)$$

**Theorem 2.1** (Canonical susceptibility). *Assume the nonzero coherent-source branch, scalar edge-center exhaustion, co-registration with the protected reserve, and the presence reading in Eqs. (2.2)–(2.3). Then*

$$\boxed{\chi_{\nu}^{\text{can}} = \lambda_{\text{collar}} = 1 - \frac{P}{24}}. \quad (2.4)$$

For  $P = 1.630968209403959$ ,

$$\boxed{\chi_{\nu}^{\text{can}} = 0.9320429912748350 \dots}. \quad (2.5)$$

*Proof.* The coherent source enters the scalar opportunity count before the protected reserve acts. Co-registration makes the local surviving fraction  $1 - \epsilon_{\mathbb{Z}_6}(y)$ . Averaging with  $w(y)$  gives  $\lambda_{\text{collar}} S_{\text{coh}}^{\text{can}}$ . The definition of canonical susceptibility gives  $\chi_{\nu}^{\text{can}} S_{\text{coh}}^{\text{can}}$ . Nondegeneracy permits cancellation of  $S_{\text{coh}}^{\text{can}}$ , and Eq. (2.3) gives the value.  $\square$

**Corollary 2.2** (Mean-count band). *If  $\epsilon_{\mathbb{Z}_6}(y)$  is interpreted only as the mean of a nonnegative integer occupancy, then*

$$\boxed{0.9320429912748350 \dots \leq \chi_{\nu}^{\text{can}} \leq 1}. \quad (2.6)$$

*Proof.* Markov's inequality gives  $1 - \epsilon \leq \Pr[N = 0] \leq 1$ . Integration and Eq. (2.3) give the band.  $\square$

The theorem excludes a zero scalar-channel coefficient on the stated nondegenerate branch. It does not exclude a zero dimensional gravitational coupling  $q_\star$ , a zero device repair charge, or a zero ambient repair gradient.

### 3 Engineering Coordinates

An engineering description may absorb an effective coherent capacity into the source coordinate:

$$S_{\text{coh}}^{\text{eng}} = N_{\text{coh}} S_{\text{coh}}^{\text{can}}. \quad (3.1)$$

Invariance of the scalar perturbation requires

$$\chi_\nu^{\text{can}} S_{\text{coh}}^{\text{can}} = \chi_\nu^{\text{eng}} S_{\text{coh}}^{\text{eng}}, \quad \boxed{\chi_\nu^{\text{eng}} = \frac{\chi_\nu^{\text{can}}}{N_{\text{coh}}}}. \quad (3.2)$$

Very small values of  $\chi_\nu^{\text{eng}}$  can therefore represent an order-one canonical coefficient. Equation (3.2) is a chart identity. It does not define  $q_\star$ , measure  $S_{\text{coh}}^{\text{can}}$ , or predict force.

### 4 Repair-Charge Coupling

The dark-sector completion promotes integer scalar repair occupation and a compact repair phase to a canonical pair  $(n, \theta)$ . On a fixed real lift of the phase, its relevant action is

$$\begin{aligned} I_{\text{RPC}} = I_{\text{Regge}} + I_{\text{SM}} \\ + \int d\tau \left[ - \sum_v V_v n_v D_\tau \theta_v - \sum_v V_v \varepsilon(n_v) - \sum_e W_e \mathcal{K}(|(d\theta)_e|) \right. \\ \left. - \sum_v V_v \theta_v \left( \mathcal{Q}_{b,v} + q_\star \chi_\nu^{\text{can}} S_{\text{coh},v}^{\text{can}} \right) \right]. \end{aligned} \quad (4.1)$$

The source and canonical signs are chosen so phase variation gives the positive-source balance in Eq. (4.3), and translation of a positive localized source gives force  $-Q_R \nabla \theta$ . A globally single-valued compact-phase coupling, the derivation of this action from finite repair dynamics, and the dimensional constants are work in progress.

The coherent repair-charge density is

$$\mathcal{Q}_{\text{coh}} = q_\star \chi_\nu^{\text{can}} S_{\text{coh}}^{\text{can}}. \quad (4.2)$$

The phase equation is

$$D_\tau n + d^\dagger j_R = \mathcal{Q}_b + q_\star \chi_\nu^{\text{can}} S_{\text{coh}}^{\text{can}}. \quad (4.3)$$

For a finite region  $\Omega$ ,

$$\frac{d}{d\tau} \sum_{v \in \Omega} V_v n_v + \sum_{e \in \partial\Omega} W_e j_{R,e} = \sum_{v \in \Omega} V_v \left( \mathcal{Q}_{b,v} + q_\star \chi_\nu^{\text{can}} S_{\text{coh},v}^{\text{can}} \right). \quad (4.4)$$

The boundary flux is part of the physical source ledger.

**Proposition 4.1** (Conditional rotor consequences). *Assume Eq. (4.1), a positive branch  $n > 0$ , and the infrared constitutive functions*

$$\varepsilon(n) = m_R n + \frac{u_R}{3} n^3, \quad \mathcal{K}(y) = \frac{\kappa_R}{3} y^3, \quad m_R, u_R, \kappa_R > 0. \quad (4.5)$$

Then:

- (i) *phase variation gives the exact finite repair-current balance Eq. (4.4);*
- (ii) *the homogeneous dilute phase has*

$$\rho_R = m_R n + \frac{u_R}{3} n^3, \quad p_R = \frac{2u_R}{3} n^3, \quad (4.6)$$

*so source-free expansion gives  $n \propto a^{-3}$ ,  $\rho_R \propto a^{-3}$ , and  $w_R \simeq 0$  when  $u_R n^2 \ll m_R$ ;*

- (iii) *if  $\mathcal{Q}_b = \beta_b \rho_b$  and  $\mathbf{a}_R = -\beta_b \nabla \theta$ , the static spherical exterior branch gives*

$$a_R = \sqrt{a_b a_0}, \quad v^4 = G M_b a_0, \quad a_0 = \frac{\beta_b^3}{4\pi G \kappa_R}; \quad (4.7)$$

- (iv) *translation of the coherent source gives Eq. (4.9) and the momentum ledger Eq. (6.2).*

*These statements are consequences of the proposed action, not derivations of that action from the finite OPH theorems.*

*Proof.* Item (i) follows by varying  $\theta$  and summing the local equation over a finite region. For item (ii), the first-order rotor Hamiltonian density is  $\varepsilon(n)$ , and its barotropic pressure is  $p_R = n\varepsilon'(n) - \varepsilon(n)$ . Source-free homogeneous continuity then gives  $na^3 = \text{constant}$ . For item (iii), spherical integration of  $\nabla \cdot (\kappa_R |\nabla \theta| \nabla \theta) = \beta_b \rho_b$  gives  $4\pi r^2 \kappa_R |\theta'| \theta' = \beta_b M_b$ ; the stated acceleration and definition of  $a_0$  give the result. Item (iv) follows by translating the source term in Eq. (4.1) and applying translation invariance to matter plus repair field.  $\square$

Define the integrated device repair charge

$$Q_R^{\text{dev}} = q_\star \chi_\nu^{\text{can}} \int_\Omega S_{\text{coh}}^{\text{can}}(x) d^3x. \quad (4.8)$$

For a prescribed external repair phase, translation of the source gives

$$F_i^{\text{coh}} = -q_\star \chi_\nu^{\text{can}} \int_\Omega S_{\text{coh}}^{\text{can}}(x) \partial_i \theta_{\text{ext}}(x) d^3x. \quad (4.9)$$

If the external gradient varies slowly across the device,

$$F_i^{\text{dev}} \simeq -Q_R^{\text{dev}} \partial_i \theta_{\text{ext}}. \quad (4.10)$$

The sign of  $Q_R^{\text{dev}}$  may select attractive or repulsive response. Positive repair energy is compatible with signed charge when  $\varepsilon(n) = \varepsilon(|n|)$ .

*Remark 4.2* (Switchable-sign boundary). The gated source in Eq. (1.1) is nonnegative under its displayed definition. With fixed  $q_\star$  and positive  $\chi_\nu^{\text{can}}$ , exchanging top and bottom source profiles does not reverse the monopole  $Q_R^{\text{dev}}$ . A switchable force sign therefore requires either a signed orientation factor derived as part of the coherent source map or a controlled reversal of  $\nabla \theta_{\text{ext}}$ . The finite channel theorem supplies neither operation.

## 5 Compact-Source Constraint

**Proposition 5.1** (No monopole from a neutral compact source). *Suppose the coherent repair source is compactly supported and satisfies*

$$\int_{\Omega} \mathcal{Q}_{\text{coh}} d^3x = 0. \quad (5.1)$$

*Then it has no repair monopole. Its exterior field begins at dipole order or higher. In a uniform external gradient, the leading forces on its positive and negative charges cancel.*

*Proof.* Integrating Eq. (4.3) in a static state makes the exterior flux equal the integrated source. Equation (5.1) sets the monopole flux to zero. The multipole expansion therefore begins beyond the monopole. A uniform external gradient couples only to total charge at leading order, which vanishes.  $\square$

A bottom-to-top contrast

$$\Delta S = S_{\text{bottom}} - S_{\text{top}} \quad (5.2)$$

can shape a dipole, select the sign of injected repair charge, couple to a nonuniform external field, or drive an alternating repair current. It does not by itself define a nonzero  $Q_R^{\text{dev}}$ . No formula proportional only to  $g^2 A_{\chi\nu} \Delta S / (4\pi G)$  is a force theorem for a closed compact device.

## 6 Momentum and Energy Closure

For the cubic condensed branch, the repair-field stress is

$$T_{ij}^R = \kappa_R |\nabla \theta| \partial_i \theta \partial_j \theta - \delta_{ij} \frac{\kappa_R}{3} |\nabla \theta|^3 + \dots \quad (6.1)$$

The exact force ledger is

$$\boxed{F_i^{\text{matter}} + \frac{dP_i^R}{dt} + \oint_{\partial\Omega} T_{ij}^R n^j dA = 0.} \quad (6.2)$$

A local repulsive response requires either nonzero device repair charge with a field tail or explicit momentum transfer through Eq. (6.2). A static, closed, repair-neutral device cannot lift itself.

Any switchable force also requires a toggle-energy ledger. Let  $E(q, s)$  be the energy at position  $q$  and internal state  $s$ , and define

$$\text{Toggle}(q) = E(q, \text{on}) - E(q, \text{off}). \quad (6.3)$$

For an on/off spatial cycle between  $q_1$  and  $q_2$ , conservation gives

$$W_{\text{cycle}} = \text{Toggle}(q_1) - \text{Toggle}(q_2). \quad (6.4)$$

A claimed position-dependent switchable force with no corresponding toggle or field-energy difference violates the closed energy ledger.

## 7 Laboratory Evidence Conditions

A force experiment has physical standing only if it supplies:

1. an instrument-independent measurement of  $S_{\text{coh}}^{\text{can}}$  or its declared engineering coordinate;
2. the dimensional calibration  $q_\star$ ;
3. a measured ambient repair gradient or a controlled emitted repair field;
4. a nonzero integrated repair charge, or a measured outgoing repair-current and momentum flux;
5. a complete toggle, maintenance-power, and field-energy ledger;
6. detuned, dummy, reversed-sign, thermal, acoustic, electromagnetic, mechanical, and center-of-mass controls;
7. a force result that follows the repair-field geometry and sign law in Eq. (4.9).
8. for any claimed switchable reversal, a derived signed source factor or an independently measured reversal of the external repair gradient.

Without these receipts, a null result does not measure  $\chi_\nu^{\text{can}}$ , and a nonzero balance signal does not establish repair charge. The lane remains outside the falsification program because the action, source calibration, and operational source observable are work in progress.

## 8 Conclusion

The canonical coherent-matter susceptibility is an order-one channel coefficient fixed by protected-reserve survival on the stated OPH branch. The repair-charge condensate completion gives that coefficient a lawful dynamical role: it weights the coherent source in the repair-current equation. The same proposed action yields exact finite continuity, a dilute dust limit, the spherical deep-galaxy scaling, repair-field stress, and a device-force law. A device force depends on the dimensional calibration, integrated repair charge, external field, and field-momentum ledger. Repulsive response is conditionally allowed. Reactionless lift from a compact neutral contrast is excluded.

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